Applicant: **Deshmukh, Amol**
Organisation: **University of Glasgow**
Funding Sought: **£732,657.46**

URF\R1\201374

Social Robots as Agents of Behaviour Change for Developing countries: Influencing Handwashing Behaviour

My research vision is to develop low-cost high impact social robotic applications for people in developing countries. Social robotics research is predominant in the developed world and having real world deployments of social robots is rare, where more than 75% of research is done in a lab without considering real world impact. On the other hand, influence and perception of social robots has barely been investigated for subjects from rural populations in developing countries which is nearly half of the population on the planet. Hence there is big scientific gap in the research of social robotics, in order bridge this gap, I have identified a specific use case where social robots could create a very high impact in developing countries, hand hygiene. In developing countries diarrhoea and respiratory infections are the two biggest causes of child mortality. By developing an intervention such as hand washing with soap, child mortality can be reduced by 40%. In this fellowship I propose a novel method to hand hygiene intervention by developing a social robot. I will develop an approach based on gamification principles and multimodal affective feedback strategies from the social robot and measure its impact by systematic empirical studies in rural schools in India. The outcomes of this research could contribute towards informing design decisions for social robotic applications that seek to address the underserved populations of the world and be applicable to other social robotics applications to effect positive behaviour change.

PRIMARY APPLICANT DETAILS

Title	Dr
Name	Amol
Surname	Deshmukh
Organisation	University of Glasgow
Website (Work)	**http://www.dcs.gla.ac.uk/~amol/**
Tel	**+447511381167**
Email (Work)	**amol.deshmukh@glasgow.ac.uk**
Address	Flat 0/1 10 Thornbank street Glasgow lanarkshire G3 8SU United Kingdom

COLLABORATOR DETAILS

Role	**Research Support**
Title	Ms
Name	Grants
Surname	Managers
Organisation	University of Glasgow
Tel	**01413301773**
Email (Work)	**rso-roysoc@glasgow.ac.uk**
Address	University Avenue Glasgow g12 8pw United Kingdom

Role	**Head of Department**
Name	Christopher
Surname	Johnson
Tel (Work)	**01413306053**
Email (Work)	**christopher.johnson@glasgow.ac.uk**
Address	University Avenue University Avenue GLASGOW Glasgow G12 8QQ United Kingdom

Role	**Nominated Referee (1)**
Title	Dr
Name	Mary Ellen
Surname	Foster
Organisation	University of Glasgow
Tel (Mobile)	**07855119957**
Email (Work)	**maryellen.foster@glasgow.ac.uk**
Address	18 Lilybank Gardens GLASGOW G12 8RZ United Kingdom

Role	**Nominated Referee (2)**
Title	Professor
Name	Ruth
Surname	Aylett
Tel	**0131 451 4189**
Email	**r.s.aylett@hw.ac.uk**
Address	Earl Mountbatten Building Heriot-Watt University Edinburgh United Kingdom

Section 1 - Eligibility Criteria

Section 2 - Contact Details

PRIMARY APPLICANT DETAILS

Title	Dr
Name	Amol
Surname	Deshmukh
Organisation	University of Glasgow
Website (Work)	**http://www.dcs.gla.ac.uk/~amol/**
Tel	**+447511381167**
Email (Work)	**amol.deshmukh@glasgow.ac.uk**
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Role	**Head of Department**
Name	Christopher
Surname	Johnson
Tel (Work)	**01413306053**
Email (Work)	**christopher.johnson@glasgow.ac.uk**
Address	University Avenue University Avenue GLASGOW Glasgow G12 8QQ United Kingdom

Role	Nominated Referee (1)
Title	Dr
Name	Mary Ellen
Surname	Foster
Organisation	University of Glasgow
Tel (Mobile)	**07855119957**
Email (Work)	**maryellen.foster@glasgow.ac.uk**
Address	18 Lilybank Gardens GLASGOW G12 8RZ United Kingdom

Role	Nominated Referee (2)
Title	Professor
Name	Ruth
Surname	Aylett
Tel	**0131 451 4189**
Email	**r.s.aylett@hw.ac.uk**
Address	Earl Mountbatten Building Heriot-Watt University Edinburgh United Kingdom

GMS ORGANISATION

Type	Organisation
Name	University of Glasgow
Email (Work)	**rso-roysoc@glasgow.ac.uk**
Address	University Avenue GLASGOW G12 8QQ United Kingdom

Section 3 - Applicant Career Summary

Title of Current Position

Please state the title of your current position.

Research Associate

Current Employer

Please enter the official organisation name of your current employer.

University of Glasgow

Current Department

Institute of Neuroscience & Psychology

Current Position Start Date

Please enter the date when your current position started.

01 April 2016

Current Position End Date

Please enter the date when your current position is expected to finish.

31 October 2020

Date of First Post-Doctoral Research Position

01 April 2016

PhD Award Date

Please enter the date that you were awarded your PhD.

01 December 2015

PhD Institution

State the name of the institution where you were awarded or are completing your PhD.

Heriot-Watt University

Personal Statement

Please provide a personal statement detailing your research career to date including prizes and achievements, and your career and research aspirations in the long term.

My career ambition is to develop social robots as agents for positive behaviour change. My key research expertise is to analyse how people interact with robots, build robotic systems that can interact effectively and influence human behaviour using user-centered design approaches. During my PhD I investigated social verbal feedback strategies to influence user acceptance of a social robot in a real-world workplace setting. I have worked on multiple European Union projects involving social robots in schools studying the effects of affective feedback in embodied robotic tutors to enhance the learning experience for children in schools and in public spaces developing non-verbal behaviours for a humanoid robot and investigating human perception towards it. I have also pioneered the first human-robot interaction study in rural populations in a developing country to study user perception and technology acceptance of a social robot.

I have published 45 (20 as first author) peer-reviewed papers and journal articles internationally and reviewed numerous journal articles and conference papers. I am the editorial board member for new open access Journal of Future Robot Life (JFRL). I was the finance chair for the IEEE RO-MAN conference in 2014 and a special session chair in 2019. I was speaker at Pint of Science, “Robots, it makes sense!”, Glasgow, 2017. I contributed in the application (specifying robotics lab equipment) for a successful grant of £6 million of UK government funding for Edinburgh Centre for Robotics in 2014. I won the best one-page statement award on “what future research is required for personal robotics and why?” at Towards Autonomous Robotic Systems conference, 2009. I built my first social robot during my Bachelor’s in Electronics and won national level robotics competitions at Asia’s largest model presentation contest, 2005. I received a scholarship for Masters from Jacobs University Bremen M.Sc. in “Smart Systems”, 2006. I have supervised 7 M.Sc student projects.

I have always pursued independent research ideas that can cause a real impact on the ground in people’s lives. Breaking the traditional research in my field, I took 2 months absence of leave and secured a funding from Scottish Informatics and Computer Science Alliance for researcher exchange funding of £3500 to travel and collaborate with Amrita University in India in 2017. I carried out an unprecedented human-robot interaction study with rural subjects in Indian villages. The research resulted in 2 publications and global media coverage such as The BBC, The Telegraph, IEEE Spectrum and a number of online media articles. I have organised the inaugural workshop on the topic “Social robots and artificial agents for developing countries: Challenges and opportunities” at 2019 RO-MAN conference. My interdisciplinary skills with electronics, computer science and the experience I have gained from developing and deploying social robots makes me an ideal candidate to undertake this research. This prestigious Royal Society fellowship will allow me to broaden my knowledge and skills and help to establish myself as a world-leading and a pioneering researcher in social robotics developing innovative solutions for rural communities in developing countries.

Applicant Career History

Please list all of your appointments since your PhD and the dates in reverse chronological order, stating if part-time (and percentage part-time) when necessary.

Research Associate
University of Glasgow, Glasgow, UK
01-04-2016-current

Research Associate
Heriot-Watt University, Edinburgh, UK
01-10-2008 - 31-03-2016

Applicant Qualifications

Please list all your qualifications in reverse chronological order.

Heriot-Watt University - 01-10-2008 - 01-12-2015
PhD (School of Mathematical and Computer Sciences)

Jacobs University Bremen gGmbH - 01-08-2006 - 01-06-2008
M.Sc - Smart Systems (Computer Science)

Savitribai Phule Pune University - 01-06-2003 - 01-06-2006
Masters in Computer Application (Computer Science)

Savitribai Phule Pune University - 01-06-2001 - 01-06-2003
Bachelors in Electronics (Electronics)

List your Key and/or Relevant Publications

Published 45 peer-reviewed papers (20 as first author) internationally

Amol Deshmukh et al., "Influencing Hand-washing Behaviour With a Social Robot: HRI Study With School Children in Rural India", The 28TH IEEE International Conference on Robot and Human Interactive Communication Communication (RO-MAN), October, 2019, New Delhi, India (accepted). [Independently led research]

Unnikrishnan R, Amol Deshmukh et al., "Design and Perception of a Social Robot to Promote Hand Washing among Children in a Rural Indian School", The 28TH IEEE International Conference on Robot and Human Interactive Communication Communication (RO-MAN), October, 2019, New Delhi, India. (accepted) [Independently led research]

Amol Deshmukh, Sooraj Krishna, Akshay Nagarajan, Vennila Vilvanathan, Sivaprasad J. V. and Bhavani Rao R., "Technology Acceptance, Sociocultural Influence and Gender Perception of Robots: A Human Robot Interaction Study with Naive Users in Rural India", The 27TH IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), August, 2018, Nanjing, China. [Independently led research] (news covered in BBC https://www.bbc.co.uk/news/av/technology-43814332/the-robot-helping-to-deliver-water-in-rural-india, The Telegraph https://www.telegraph.co.uk/global-health/climate-and-people/robots-coming-fetch-water/, IEEE Spectrum https://spectrum.ieee.org/automaton/robotics/robotics-hardware /robot-research-in-the-wild-water-transport-in-rural-india)

Amol Deshmukh et al., "Shaping Robot Gestures to Shape Users' Perception: The Effect of Amplitude and Speed on Godspeed Ratings", In 6th International Conference on Human-Agent Interaction, December

15–18, 2018, Southampton, United Kingdom. ACM, New York, NY, USA

Amol Deshmukh et al., "Social Impact of recharging activity in long-term HRI and verbal strategies to manage user expectations during recharge", Journal Frontiers in Robotics and AI- Humanoid Robotics, 2018

Helen Hastie, Katrin Lohan, Amol Deshmukh, Frank Broz and Ruth Aylett, "The Interaction Between Voice and Appearance in the Embodiment of a Robot Tutor", The Ninth International Conference on Social Robotics November, 2017, Tsukuba, Japan

Amol Deshmukh et al., "Modulating the Non-Verbal Social Signals of a Humanoid Robot", 19th ACM International Conference on Multimodal Interaction, November 2017, Glasgow, Scotland, United Kingdom

Amol Deshmukh et al., "Perceptual Control Theory for Engagement and Disengagement of Users in Public Spaces", ACM/IEEE, International Conference on Human-Robot Interaction, Vienna, Austria, 2017

Amol Deshmukh et al., "How Expressiveness of a Robotic Tutor is Perceived by Children in a Learning Environment", ACM/IEEE, International Conference on Human-Robot Interaction, Christchurch, New Zealand, March 2016

Mary Ellen Foster, Amol Deshmukh et al., "How does Affective robot feedback influence learner experience in a real-world treasure hunt?" In Proceedings of First International Workshop on Educational Robots (WONDER), 2015

Mary Ellen Foster, Amol Deshmukh et al., "Influencing the Learning Experience Through Affective Agent Feedback in a Real-World Treasure Hunt." In Proceedings of the 14th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2015), pages 1711-1712, Istanbul, May 2015

Amol Deshmukhet et al., "Towards Empathic Artificial Tutors", Proceedings of the 8th ACM/IEEE international conference on Human-robot interaction, March 2013, Tokyo Japan

Applicant Research Funding

Please list all your current and previous research funding in reverse chronological order.

Lord Kelvin Adam Smith (LKAS) Fellowship funding £100,000 to supplement this application, 2019.

Travel grant of £3,500 from SICSA - Scottish Informatics and Computer Science Alliance for post doctoral and Early Career Researcher Exchange, 2017.

Contributed in defining and procuring robotics lab equipment for Heriot-Watt University, Edinburgh Centre for Robotics, 2014 (successful grant £6 million of UK government funding)

Section 4 - Research Proposal

Start Date

Please enter the proposed start date of the project.

01 October 2020

End Date

Please enter the proposed end date of the project.

30 September 2025

Project Title

Please give the full title of your proposed project.

Social Robots as Agents of Behaviour Change for Developing countries: Influencing Handwashing Behaviour

Keywords

Provide up to 5-10 key words that could be used to describe your research, which will aid with identifying the most appropriate peer reviewers.

Human-robot interaction, social robotics, handwashing, persuasive technologies, gamification, affective computing

Subject Group

Computer Science

Subject Area

- ☑ Computer Engineering (incl. software)
- ☑ Control (incl. robotics)
- ☑ Experimental Psychology
- ☑ Development and Control of Behaviour
- ☑ Artificial Intelligence
- ☑ Machine Learning
- ☑ Vision
- ☑ Human-computer interaction

Abstract

Please provide a scientific summary of your proposed project. This should be a summary of your research proposal, briefly outlining the background and summarising the aims of your project.

This fellowship aims to develop social robots as agents for persuasive technologies that can autonomously interact with people and effect positive change on their behaviour. Since most of the social robotics research is carried out in developed countries, my research aims to (i) fill the scientific gap in our understanding of how people in developing countries from rural backgrounds interact with social robots and (ii) develop the interaction strategies for social a robot to effect positive change on their behaviour. This research will be carried in context to a very important issue in public health in developing countries, hand hygiene compliance. Handwashing With Soap (HWWS) could save lives, reduce germ transmission, save public health costs and decrease infection-related absenteeism in schools. Existing practices for influencing handwashing behaviour in developing countries face three major challenges (i) they are very resource intensive, (ii) complex to measure/monitor compliance at scale and (iii) they rarely are sustainable over long-term. Hence there is need for a multi-disciplinary and innovative approach for influencing sustainable handwashing behaviour change. However, none of the previous hand hygiene interventions in developing countries have considered the use of interactive technology using social robots. Social robots can, for example, encourage, remind, educate, give real-time feedback, observe and monitor hand washing behaviour. The key research questions to be addressed in this fellowship are (i) what interaction design strategies (verbal and non-verbal) are the most effective in influencing handwashing behaviour?, (ii) how can social robots achieve sustainable handwashing behaviour change? In order to achieve the goal of this research, there is a need to develop the autonomous capabilities, which is a significant scientific and technical step in this research. This research will develop an approach based on gamification principles and

multimodal affective feedback strategies from the social robot. The research methodology will involve: (i) a multi-disciplinary approach combining behavioural science and strategies of human factors engineering into the behaviour of the social robot, (ii) user-centered iterative design approach within the target population (iii) empirical studies to investigate the influence of a social robot on handwashing behaviour. I will initially target India which has the world's largest rural population of 850 million, with only 15% practising HWWS, and then diversifying to other countries. The annual net costs to India from not washing hands are estimated at US $ 23 billion, so a national handwashing program in India could provide an estimated 92-fold return on investment.

Lay Summary

My research vision is to develop low-cost high impact social robotic applications for people in developing countries. Social robotics research is predominant in the developed world and having real world deployments of social robots is rare, where more than 75% of research is done in a lab without considering real world impact. On the other hand, influence and perception of social robots has barely been investigated for subjects from rural populations in developing countries which is nearly half of the population on the planet. Hence there is big scientific gap in the research of social robotics, in order bridge this gap, I have identified a specific use case where social robots could create a very high impact in developing countries, hand hygiene. In developing countries diarrhoea and respiratory infections are the two biggest causes of child mortality. By developing an intervention such as hand washing with soap, child mortality can be reduced by 40%. In this fellowship I propose a novel method to hand hygiene intervention by developing a social robot. I will develop an approach based on gamification principles and multimodal affective feedback strategies from the social robot and measure its impact by systematic empirical studies in rural schools in India. The outcomes of this research could contribute towards informing design decisions for social robotic applications that seek to address the underserved populations of the world and be applicable to other social robotics applications to effect positive behaviour change.

Research Proposal

You can either detail your research proposal within this application form, or alternatively upload it as a PDF file. PDF files must be no longer than 3 sides of A4, portrait orientation, be titled and the text size cannot be smaller than Arial size 10.

Your research proposal should provide a general description of the proposed research to be carried out. This should include, but is not limited to, the following:

- **clear specification of the context, research objectives and expected outputs of the proposed study;**
- **description of the methodology to be used, including data analysis;**
- **an indication of the milestones and timescales**

Please specify your preferred method for detailing your research proposal.

⦿ PDF upload

Research Proposal

Please upload your proposal as a PDF file. PDF files must be no longer than 3 sides of A4, portrait orientation, be titled and the text size cannot be smaller than Arial size 10.

URF_Proposal
02/09/2019
15:49:04
pdf 449.55 KB

If you have named collaborators in your research proposal please provide letters of support.

Support-letter-Amrita

02/09/2019

12:18:05

pdf 1.69 MB

Host Organisation

Please select the organisation where the award will be held from the drop down list below.

University of Glasgow

Host Department

Computing Science

Justification for Choice of Host Organisation

This fellowship will take place in the School of Computing Science (SoCS) at the University of Glasgow (UOG). UOG, is 55th in the QS World Ranking 2014 and has annual research income of more than £181M. The School of Computing Science (SoCS) is rated in the UK's top 10 for research, top in Scotland for research impact (REF 2014), 6th in the UK for research intensity (REF 2014), top 100 in the World (QS World University Subject Rankings 2018). Current research funding amounts to more than £6M in SoCS. I will work within the Social Robotics group at the school of Computing Science which is collaborative and interdisciplinary, working closely with other schools including Psychology and the Institute of Health and Wellbeing. The SoCS has recently been awarded a Centre for Doctoral Training (CDT) together with the Institute for Neuroscience and Psychology to train a cohort of 50 PhD students on Socially Intelligent Artificial Agents. I will have the opportunity to co-supervise PhD students from CDT. This research on the use of social robots for public health in the developing world fits within the GIST (Glasgow Interactive Systems) group at computing science department. GIST aims to create and use novel, interactive systems to better understand, entertain, protect and support humans in their everyday lives. I will receive guidance and advice from Professor Alessandro Vinciarelli an expert in the domain of modelling, analysis and synthesis of nonverbal human-machine interactions and Dr Mary Ellen Foster from the school, an expert in real-world deployment of social robots especially on verbal and non-verbal feedback strategies. In addition have two mentors from UOG, from Institute of Health and Wellbeing Prof Sally Wyke, Deputy Director and Dr Katie Robb. They are experts in applying interdisciplinary social science theory, developing and evaluating interventions for health and wellbeing appropriated to context and culture. This research will also benefit from connections with the Glasgow Centre for International Development (GCID), a pan-University network that aims to create an interdisciplinary, collaborative and supportive environment for researchers working with collaborators in low and middle income countries. UOG was ranked 9th among UK universities for GCRF/Newton funding secured, is leading holder in Scotland for international development research income having more RCUK GCRF funded research than any other Scottish institution. I have secured an additional £100,000 funding support from UOG's Lord Kelvin Adam Smith (LKAS) Fellowship funds for a PhD student and buy additional equipment, video annotations, travel support for conducting field studies to supplement this proposed research. I will also be enrolled on UOG's early career development programme which aims to accelerate the career progression of young academics. UOG provides an excellent research environment to carry out my research in the most effective manner.

Outline of Data Management and Data Sharing Plan

If the proposed research will generate data that is of significant value to the research community, then please provide details of your data management and sharing plan.

The data generated from this research will comprise of questionnaires in excel files, image and video data from field studies. All data will be made anonymous conforming to privacy and ethical approval by Amrita

University in India and University of Glasgow ethics committee. Informed and written consent will be obtained from the school authorities and children's parents before recording, and only research staff would have access to the data. I will use robust security measures to protect participants' data, such as encryption, and store data securely on the University's servers for a maximum period of five years following the project. I will use automated software to pixelate faces (using Adobe After Effects) and remove audio on recorded footage after analysis has been performed. Data will be stored on encrypted hard drive with password protection. All data storage and provision will be compliant with GDPR procedures as implemented at UOG. The video and image data collected from handwashing (no faces, hand movements only) will be valuable to train machine learning algorithms on hand gesture recognition for the wider community and will be made available on https://www.openml.org/.

Section 5 - Use of Animals in Research

Does your proposal involve the use of animals or animal tissue?

⦿ No

Section 6 - Use of Human Patients and Tissue

Does your proposal involve the use of human patients or tissue?

⦿ No

Section 7 - ODA Compliance

Please indicate if you think your proposed research meets the remit of the Global Challenges Research Fund.

⦿ Yes

How is your proposal directly and primarily relevant to the development problems of these countries?

Handwashing with soap is included in the United Nations (UN) sustainable development goals Target 6.2 for sanitation and hygiene. Every year, nearly 443 million school days are lost due to water and sanitation related diseases in developing countries. My proposed research idea reflects a low-cost method with the potential to increase hand hygiene compliance, which could reduce germ transmission, save public health costs and decrease infection-related absenteeism in schools. My research if scaled could also contribute towards meeting UN's sustainable development goals targets around sanitation, hygiene, education, nutrition and child survival in developing countries. Social robots could have a great potential to serve a supportive role in large scale health interventions both as active agents of behaviour change as well as objective assessors of various indicators of health and hygiene. My preliminary research has shown that it is important to understand the scientific and interaction design challenges of introducing robotic solutions in a real life rural setting. The UN's sustainable development goals also relate to bridging the digital divide, making technology more accessible to all users rather than a privileged few in developed countries. My research helps identifying a valuable and viable use case for social robots in rural populations.

How do you expect that the outcome of your proposed activities will promote the economic development and welfare of a country or countries on the DAC list?

India, a lower middle income country, has the world's largest rural population of approximately 850 million, about 30% under 14 years of age (270 million), and only 15% of the entire population practices handwashing with soap. Townsend et al. in a study, (https://doi.org/10.1111/tmi.12808, 2017) reports that the DALYs (Disability Adjusted Life Years, calculated as the sum of the Years of Life Lost due to premature

mortality and the years lost due to Disability) due to diarrhoea in India amounted to 29.77 million, 14.63 million of which were for infants under 5 years of age and 3.14 million for children 5–14 years of age. The economic loss to India from the 29.77 million DALYs due to diarrhoea is estimated at US$ 47.157 billion (US$ 23.169 billion for children under 5 years and US$ 8.955 billion for children 5–14 years). Since the target age group for the proposed intervention is primary school children aged between 5-11, even with a rough estimate considering the above figures the economic loss for that age group is huge. The population contracting diarrhoea due to not washing their hands is estimated at 90.4% and handwashing with soap can reduce the risks by 40% (preventable costs upto US$ 3-4 billion for age group 5-14). We are not targeting the entire population in our research, but the potential cost savings if the intervention is sustainable can produce huge economic savings. My research idea also compliments the national campaign launched by Indian government "Swachh Bharat – Swachh Vidyalaya" (Clean India – Clean Schools) working to improve water, sanitation and hygiene (WASH) services and practices across India. I shall pursue to take this technology to other countries, I am an affiliate member of Sustainable Futures in Africa (SFA) network which spans across countries like Uganda, Botswana, Nigeria and Malawi.

DAC Country list

- ☑ India

Section 8 - Financial Details

Please define the proposed budget for your Fellowship in the table below, ensuring that you provide values at 100%:

Period Group	Item Type	Item	Field	£
2020 - 2021	Basic Salary	Basic Salary	Cost	£40,681.46
	London Weighting Allowance	London Weighting	Cost	£0.00
	On costs	On costs	Cost	£14,975.00
	Consumables (inc. fieldwork)	Consumables	Cost	£3,000.00
	Other research expenses	Other research expenses	Cost	£2,000.00
	Animal purchases	Animal purchases	Cost	£0.00
	Animal maintenance	Animal maintenance	Cost	£0.00
	Travel (inc. subsistence)	Travel (inc. subsistence)	Cost	£5,000.00
	Equipment	Equipment	Cost	£3,000.00
	Indirect costs	Indirect costs	Cost	£56,058.00
	Estate costs	Estate costs	Cost	£16,509.00
	Other costs	Other costs	Cost	£0.00
2020 - 2021 Total			**Cost**	**£141,223.46**
2021 - 2022	Basic Salary	Basic Salary	Cost	£41,902.00
	London Weighting Allowance	London Weighting	Cost	£0.00
	On costs	On costs	Cost	£15,424.00
	Consumables (inc. fieldwork)	Consumables	Cost	£1,000.00
	Other research expenses	Other research expenses	Cost	£2,000.00
	Animal purchases	Animal purchases	Cost	£0.00

	Animal maintenance	Animal maintenance	Cost	£0.00
	Travel (inc. subsistence)	Travel (inc. subsistence)	Cost	£5,000.00
	Equipment	Equipment	Cost	£3,000.00
	Indirect costs	Indirect costs	Cost	£56,058.00
	Estate costs	Estate costs	Cost	£16,509.00
	Other costs	Other costs	Cost	£0.00
2021 - 2022 Total			**Cost**	**£140,893.00**
2022 - 2023	Basic Salary	Basic Salary	Cost	£43,159.00
	London Weighting Allowance	London Weighting	Cost	£0.00
	On costs	On costs	Cost	£15,887.00
	Consumables (inc. fieldwork)	Consumables	Cost	£2,000.00
	Other research expenses	Other research expenses	Cost	£1,000.00
	Animal purchases	Animal purchases	Cost	£0.00
	Animal maintenance	Animal maintenance	Cost	£0.00
	Travel (inc. subsistence)	Travel (inc. subsistence)	Cost	£5,000.00
	Equipment	Equipment	Cost	£3,000.00
	Indirect costs	Indirect costs	Cost	£56,058.00
	Estate costs	Estate costs	Cost	£16,509.00
	Other costs	Other costs	Cost	£0.00
2022 - 2023 Total			**Cost**	**£142,613.00**
2023 - 2024	Basic Salary	Basic Salary	Cost	£44,454.00
	London Weighting Allowance	London Weighting	Cost	£0.00
	On costs	On costs	Cost	£16,363.00
	Consumables (inc. fieldwork)	Consumables	Cost	£1,000.00
	Other research expenses	Other research expenses	Cost	£2,000.00
	Animal purchases	Animal purchases	Cost	£0.00
	Animal maintenance	Animal maintenance	Cost	£0.00
	Travel (inc. subsistence)	Travel (inc. subsistence)	Cost	£6,000.00
	Equipment	Equipment	Cost	£2,000.00
	Indirect costs	Indirect costs	Cost	£56,058.00
	Estate costs	Estate costs	Cost	£16,509.00
	Other costs	Other costs	Cost	£0.00
2023 - 2024 Total			**Cost**	**£144,384.00**
2024 - 2025	Basic Salary	Basic Salary	Cost	£45,787.00
	London Weighting Allowance	London Weighting	Cost	£0.00
	On costs	On costs	Cost	£16,854.00
	Consumables (inc. fieldwork)	Consumables	Cost	£1,000.00
	Other research expenses	Other research expenses	Cost	£1,000.00
	Animal purchases	Animal purchases	Cost	£0.00
	Animal maintenance	Animal maintenance	Cost	£0.00
	Travel (inc. subsistence)	Travel (inc. subsistence)	Cost	£6,000.00

	Equipment	Equipment	Cost	£1,000.00
	Indirect costs	Indirect costs	Cost	£56,058.00
	Estate costs	Estate costs	Cost	£16,509.00
	Other costs	Other costs	Cost	£0.00
2024 - 2025 Total			**Cost**	**£144,208.00**

Total	Basic Salary	Basic Salary	Cost	£215,983.46
	London Weighting Allowance	London Weighting	Cost	£0.00
	On costs	On costs	Cost	£79,503.00
	Consumables (inc. fieldwork)	Consumables	Cost	£8,000.00
	Other research expenses	Other research expenses	Cost	£8,000.00
	Animal purchases	Animal purchases	Cost	£0.00
	Animal maintenance	Animal maintenance	Cost	£0.00
	Travel (inc. subsistence)	Travel (inc. subsistence)	Cost	£27,000.00
	Equipment	Equipment	Cost	£12,000.00
	Indirect costs	Indirect costs	Cost	£280,290.00
	Estate costs	Estate costs	Cost	£82,545.00
	Other costs	Other costs	Cost	£0.00
Total			**Cost**	**£713,321.46**

Justification for Research Expenses

1) Requesting £8,000 for Consumables (inc. fieldwork), i anticipate costs associated with conducting field studies, mainly for year 1, 3 and 4 for building prototypes, video annotations/editing software, 3d printing.

2) Requesting £8,000 for Other expenses: £4,000 for two summer student interns for year 2 and 4 to carry out video annotations, transcribe and translate interviews on data collected from field studies also some coding tasks are anticipated. Public engagements such as running workshops at schools in India and Glasgow for hand hygiene awareness games using the social robot. Also to attend International Computer Vision Summer School in 2021.

3) Requesting £27,000, Travel (inc. subsistence) as this research will require me to travel multiple times to India to conduct field studies and conference travel. I plan to spend 2 months in India during field studies in Year 1, 3 and 4. The travel costs are expected to be higher due to remote locations where field studies will be conducted. Also having student volunteers to assist with the field studies is necessary where you need volunteers to speak the local language in order to conduct interviews with the participants. From my previous experience doing field studies in small villages, researchers need to stay in remote accommodations for a few days in order to complete the study.

4) Equipment £12,000, Electronic components for building the prototype and other hardware required for the system. Powerful computer with GPU for vision processing algorithms, cameras, hard drives to store video data, wireless motion sensors.

5) PhD Studentship (£19,336) for the first year, the remaining (three years, £59,849) costs associated with the studentship, (fees at UK level and stipend) will be covered by the University LKAS (Lord Kevin Adam Smith) leadership funds. Further £8,000 have been secured through the LKAS fund to support the a PhD student at Amrita University for 4 years.

Section 9 - Applicant Declaration

Social Robots as Agents of Behaviour Change for Developing Countries: Influencing Handwashing Behaviour

1 Overview:

In this fellowship I will develop a low-cost (≈ US $ 250) social robot to create a step change in the efficacy of influencing Handwashing With Soap (HWWS). HWWS is one of the most cost-effective public health interventions in reducing the burden of global infectious diseases by 40%.[1] To carry out this research effectively, I have formed a collaboration with Amrita University and their team in India,[2] who are engaged in technology interventions and community development activities in over 142 villages in India. Handwashing interventions in schools are an effective way of reaching children and teaching them the habit of handwashing at a young age.[3] Given the ambition and the risks in this research vision, I conducted an independent research pilot project in collaboration with Amrita University in India. I led a team of 4 researchers and developed a proof-of-concept a low-cost (≈ US $ 100) social robot which was remotely controlled (not autonomous) in a rural village school in India (see figure 1). The pilot study showed encouraging results: a staggering 40% increased level of HWWS and with a better handwashing technique.[4,5] However, further research is needed to establish a solid empirical foundation to inform interaction strategies and to develop the autonomous technology required for the social robot to influence the target subjects into adopting the decision to perform handwashing behaviour.

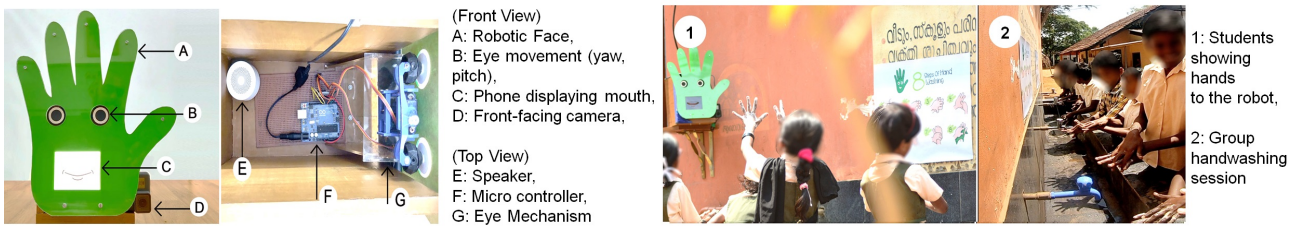

Figure 1: Image Left: Prototype Robot Design; Image Right: The wall mounted social robot near the water tap encouraged around 100 children to wash their hands via a poster on the wall,[4,5] March 2019.

Related work: In developing countries hand hygiene interventions mostly focus on educating people about health, germs and disease often using tools such as games, radio/TV campaigns, books, songs, posters, leaflets and charts.[6] Such methods can be very resource intensive and cumbersome to monitor at scale, measurements are unreliable due to self-reporting and these approaches rarely result in positive, sustained behaviour change.[7] In developed countries, hand hygiene interventions have investigated the use of electronic message boards,[8] visual[9] and audio[10] prompts as a reminder to perform hand hygiene,[11] emoticon-based feedback[12] and vision-based system for tracking people's activity in hospitals[13] with some degree of success. The state-of-the-art technology developed by SureWash (www.surewash.com) uses computer vision technology for hand hygiene training to hospital staff, costs about £30k, it is bulky, works only in indoor environments and has limited interaction capabilities. Summing up, most technology based hand hygiene interventions are expensive, they have been performed in developed countries in indoor environments and none have explored the use of social robots. In regards to social robots as persuasive applications, use of verbal[14] and non-verbal behaviour such as eye gaze[15] (i.e. looking at a participants' face), has shown positive results. However, whether the persuasive effects of the robot's verbal and non-verbal cues will influence decision making in real-world scenario such as handwashing is still undiscovered.

Approach and Objectives: The approach will focus on investigating the use of gamification which has shown positive results in technology based education[16] and health related contexts.[17,18] Gamification is the process of including game like elements that could motivate and influence human behaviour to enhance learning. However there is no evidence to suggest that gamification strategies are effective to increase hand hygiene compliance for children from rural backgrounds in developing countries using social robots. My previous work with affective feedback has shown promising results with social robots as tutors.[19] However, more empirical research is required to determine whether both extrinsic and intrinsic motivation to wash hands can be influenced by gamification combined with affective feedback strategies. The main objectives of this research are (i) to evaluate which interaction and design strategies are effective to engage and influence handwashing with children in schools, (ii) to develop the autonomous capabilities for the social robot for perception and synthesis of behaviours, and (iii) conduct field studies to measure the impact and sustainability of intervention. I aim to carry out interventions using the social robot in 10-15 schools overall (target reach approximately 2,000 children, aged 5-11 years) across different states in India during the research period. My plan for the Fellowship is distributed into 4 workpackages (WPs), the outcome of each WP informs the design strategy for the next WP. There is also possibility for WPs to run in parallel to some extent as some activities can be carried out independently. I expect to publish the findings from the outcomes generated from each WPs in relevant and leading conferences and journals.

2 Programme of Work and Methodology

2.1 WP1- Evaluation of interaction and design strategies for the social robot, *Year 1*

The year 1 goal is to evaluate the interaction design strategy necessary to address the scientific challenges in the context of this research by means of an informed design process. I will conduct user studies by using Wizard of Oz (WoZ) which is an important technique in social robotics research to get initial impressions about interaction designs.[20] The aim of the study will be to investigate the following: (1) how gamification combined with affective feedback strategies (emotions) expressed through verbal and non-verbal behaviour from the robot alters its persuasiveness and how it may be perceived by participants and (2) what important perceptual cues that need to be recognised and handled. Examining these characteristics can be useful while incorporating the findings into the design of autonomous system (WP2).

Some studies have shown the use of emotional drivers significantly increased HWWS, for example disgust (the desire to avoid and remove contamination) using an animated film/video.[6,21] However, the use of emotional drivers using affective feedback by means of social robots in hand hygiene context has not be explored before. For affective expressiveness, the social robot will also have eye movement to track users and display emotions e.g. happy, sadness, disgust. A screen using a mobile phone to display graphical mouth to display smiles, frowns and hand hygiene educational videos and a loud speaker for sounds and speech synthesis. Gamification strategies can include for example, if the children wash their hands effectively, a time based graphics will be shown on the screen (e.g. balloon bursts or a rocket fires). Verbal and non-verbal feedback strategies using social reinforcement[14] (e.g., the robot expressing social praise by saying “You are doing a great job, keep it up” while showing a happy facial expression and sounds). A parallel-design, randomised cluster experiment will be performed using the above approach near the handwashing station in 3 rural schools in India.

Measurements: Mixed research methods i.e. qualitative and quantitative data will be collected using video recordings of the interaction and questionnaires. The video data will be analysed to determine behaviour cues and social signals from the users such as proxemics (i.e. coming close to the robot), facial expressions (i.e., smiles), oculesic behaviour i.e. eye contact (frequency, duration), speech utterances, and body movements (gestures, head pose) to measure the degree of engagement with the robot. Annotations using BORIS, an event logging software for video/audio coding will be performed.[22] Videos of participants washing hands will also be recorded from a top camera. In addition, qualitative feedback will be obtained from focus groups using questionnaires to get detailed perception about the design features of the robot using Smileymeter, a visual likert scale used for user studies in child computer interaction.[23] The questionnaires will capture perceptions about appearance, verbal and non-verbal behaviours of the robot. The findings from these studies will be consolidated to define interaction and design strategies for the social robot.

2.2 WP2 - Development of autonomous capabilities for the social robot, *Year 2, 3*

WP2 will focus on developing the autonomous capabilities for the social robot informed by the findings from WP1. This WP is sub-divided into 3 main components, physical design, perception and synthesis.

Physical design: The appearance of the robot will be refined following feedback from focus group in WP1 and the new design will be fabricated using 3D printing facilities available at Glasgow University. The robot will be powered using cost-effective computers such as Raspberry Pi and smart phones for screen on the robot to display graphical/educational contents relevant to handwashing.

Perception: The technical challenge is to recognise hand gestures and behaviour cues in outdoor light sensitive and unstructured environment due to diverse physical setups of handwashing stations in different schools. To determine the quality of handwashing, motion detection using background subtraction algorithm within a region tightly cropped into the area over the water tap will be developed. Evaluation will be carried out using the video data gathered from WP1 to check the feasibility for training machine learning algorithms and facial analysis software to asses recognition of behaviour cues (e.g. body orientation, smile detector, head pose using k-means clustering algorithm) and quality of handwashing, for example the 6 steps of the WHO hand hygiene technique (hand movement).[24] I am collaborating with Prof. Gerard Lacey, Associate Professor in Graphics & Vision at Trinity College Dublin and the CEO & Co-Founder of SureWash (www.surewash.com). Prof. Lacey will be a scientific advisor for this research in the area of computer vision technology and system performance evaluation. I will have a PhD student working on the vision component of the system. The overall perception system setup will combine computer vision and wireless infrared motion sensors register children passing by the toilet, approaching the handwashing station.

Synthesis: I will incorporate the findings from WP1 into interaction strategies for the social robot. Both verbal and non-verbal strategies such as using speech, sounds, eye tracking and graphical mouth using the screen on the robot will be developed. The verbal feedback will be delivered in the local regional language the children can understand. Child-like and gender neutral voices are most effective in child-robot interaction so we will incorporate that in our speech design.[25] As there is lack of text-to-speech synthesis systems for Indian regional languages, human voice recordings can be manipulated by shifting the pitch of the sounds file to resemble that of a child whose gender is not apparent in the voice to avoid gender and age perception bias.

2.3 WP3 Field studies, *Year 3, 4*

The aim of the field studies will be to investigate (i) influence, (ii) learning transfer and (iii) sustainability of the intervention. The first phase of WP3 will involve pilot testing the robot developed in WP2 in 2 rural schools to evaluate the system performance (perception accuracy) and address technical defects of the social robot. Following an iterative design methodology the interaction strategy will be fine tuned following the results from the pilot test. The final system will be incorporated into 3 robots for deployment. Extensive field studies will be carried out in 10-15 rural schools across different states in India to capture the cultural diversity in regards to perception and influence of the social robot. The plan to is to carry out staggered interventions to evaluate sustainability of the handwashing behaviour with stepped-wedge design methodology.[26] This methodology could help to mitigate novelty effects imposed by exposure of the intervention due to repeated interactions. In the initial stage to investigate the influence of the social robot, a 4 week intervention will be carried in rural schools. Each study will be divided into 3 phases, (i) pre-robot intervention (baseline, no robot, one week) (ii) robot intervention when the robot will be placed at the handwashing station (2 weeks) (iii) post-robot when the robot is removed (one week). In all 3 phases handwashing video data will collected by placing an overhead (capturing handwashing) and front (capturing interaction) cameras near the handwashing station and pre-post questionnaires will be used as described in WP1.

Analysis: Learning transfer will be measured by analysing the handwashing behaviour by manually annotating each handwashing session using BORIS software and system logs. Objective evaluation on the quality of handwashing will be measured by how many steps the children carried out of the 6 steps of handwashing[24] and the duration, frequency of handwashing during critical moments such as before meals and after use of toilet. The influence of the social robot will be analysed by measuring verbal and non-verbal behaviours that positively correlate with desirable action from the participant for e.g. responding to the instructions from the robot and their reaction time. The synthesis of the results will also benefit from taxonomy of specific techniques to change behaviour.[27] In the final stage, video data from schools will be collected without the robot 6 months after the initial study to measure if the intervention was sustainable. Effort will be made to collect data from schools to measure the infection-related absenteeism pre and post studies to analyse sustainability.

2.4 WP4 - Consolidating results, dissemination and exploring other use cases, *Year 5*

Meta analysis of the results from all field studies will be consolidated into 3 journal publications such as International Journal for Social robotics, Computers & Education and Open Access journals (Lancet Global Health). The main outcome from the meta analysis could provide new understanding how people rationalise their behaviour and be useful in designing appropriate interaction strategies using social robots in other scenarios as agents for positive behaviour change. Publications will be happening throughout the fellowship, however results will be further disseminated through social media channels, press releases, local meetings, workshops in local schools in Glasgow, seminars and events such as Global Handwashing Day and science festivals in Edinburgh and Glasgow. During this phase having connected to rural communities other use cases for social robots will be investigated by means of unstructured interviews with stakeholders. The areas where social robots can be influential will be centred around UN's sustainable goals but not limited to topics, Good Health & Well-being, Quality Education, Addiction awareness, Gender Equality, Clean Water and Sanitation. After the fellowship I will seek to lead a multidisciplinary research groups to advance social robotics research for developing countries.

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University Research Fellowship Team
The Royal Society
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3 September 2019

Dear Sir/Madam,

RE: Social Robots as Agents of Behaviour Change for Developing countries: Influencing Handwashing Behaviour: Dr. Amol Deshmukh

On behalf of AMMACHI Labs at Amrita Vishwa Vidyapeetham University, India we are pleased to express our support of Dr. Deshmukh's application entitled "**Social Robots as Agents of Behaviour Change for Developing countries: Influencing Handwashing Behaviour**", submitted under The Royal Society URF Fellowship program.

In the proposed fellowship, Dr. Deshmukh will be investigating the influence of handwashing behaviour using social robots with rural subjects in India. Apart from being the Director of AMMACHI Labs, I also serve as the UNESCO Chair in Gender Equality & Women's Empowerment and head the R&D activities that focuses on Human-Computer Interaction (HCI), Robotics and Automation technologies for promoting gender equality, quality STEM education, skill development, disaster response, medical assistance and rehabilitation. I am the Principal Investigator for several projects at Amrita, funded by organizations such as the United Nations Democracy Fund (UNDEF), Internet.org, NORRAG, the Ministry of Human Resources & Development (MHRD) and the National Skill Development Corporation (NSDC). AMMACHI Labs brings an interdisciplinary approach to addressing societal challenges and as a Center of Excellence, we envision a world where technology is designed and used to empower people. We create innovative educational tools and skill development solutions to help uplift rural communities. We are engaged in technology interventions and community development activities in over 142 villages in India, and we believe Dr. Deshmukh's research will supplement these outreach activities. Our group is also involved with school health programs to spread handwashing awareness and promotion where we have trained hundreds of children and adults in rural areas in safe handwashing practices. We also have an extensive network of social workers working with schools and after school tuition centers in rural areas across 21 states with a potential to work with more than a thousand children, which is ideal for Dr. Deshmukh to conduct his research studies. I am pleased to extend my support for Dr. Deshmukh at Amrita University.

Dr. Deshmukh's proposal is timely and extremely relevant for the Indian context as India has a dismal 15% handwashing prevalence rate. His innovative idea of using low-cost social robots for influencing handwashing behaviour is relevant to the national campaign launched by the

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Indian government "Swachh Bharat – Swachh Vidyalaya" (Clean India – Clean Schools) working to improve water, sanitation and hygiene (WASH) services and practices across India. To the best of our knowledge, there is no research relevant to using social robots for rural populations in India, hence his research idea has the potential to create a positive impact on the wider community. Dr. Deshmukh will also have the opportunity to receive advice in the design of qualitative and quantitative methodologies for carrying out field studies in rural villages supplemented with researchers from the AMMCHI labs team. As the primary subjects of this research will be children, Dr. Sidney Strauss, an expert in child development and education and Dr. Meltem Akoyak-Yildiz from pedagogic and education sciences from Amrita University will provide guidance for his research. We were pleased to have him as a visiting researcher in 2017 to study the technology perception of rural subjects with a use-case of water carrying social robot which resulted in 2 publications and global media publicity. Recently he also led a team of researchers from AMMACHI labs in March 2019 to carry out a pilot study with a social robot to influence handwashing behaviour in a rural school in Kerala, India and the study has resulted in 2 publications. His fellowship proposal is very much in line with our strategic vision and we would be able to provide support for his field studies and enable him to carry out this project in the most effective way possible.

I have discussed with Dr. Deshmukh the various ways we can collaborate in his fellowship -

1. Guidance on the design of qualitative and quantitative methodologies for carrying out field studies in rural villages supplemented with researchers from Ammachilab team;
2. Facilitate Dr. Deshmukh to public engagement activities that we participate in raising hand hygiene awareness in villages;
3. Co-supervision of the Ph.D. student whose studentship he managed to secure through the LKAS Leadership Fellowship Fund from the University of Glasgow;
4. Access to our laboratories and maker space for building prototypes and support during the experimental phases of the project in India;
5. Assist Dr. Deshmukh in conducting research studies in the villages with our team of local language experts and field workers;
6. Facilitate and guide Dr. Deshmukh in getting approval from the university research ethics committee for all his research studies.

We look forward to Dr. Deshmukh's receipt of this prestigious fellowship and the opportunity. Please contact me if I can provide any additional information.

Sincerely,

Bhavani B

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